

ChemRisk-AI: Mechanism-Aware Mitigation Decision Support

A weak-supervision framework for chemical incompatibility screening, direction-aware RTD recommendation, and tank-level safeguard decisions

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Revision: Directionality-Aware v4 | Updated July 2026

Reader Orientation: What This Project Is About

This project sits at the intersection of chemical process engineering, process safety, and applied machine learning. It does not try to replace chemical experts or perform first-principles reaction modeling. Instead, it builds a decision-support system that helps engineers decide which safeguards are appropriate when two incompatible chemicals could be accidentally mixed.

The key engineering decision in the Phase 1 MVP is whether a temperature sensor, called an RTD, is a useful and creditable secondary safeguard for a given chemical mixing scenario and, ultimately, for the storage tank that holds one of those chemicals. The design now explicitly separates pair-level incompatibility from direction-specific tank scenarios, because A entering Tank B can differ from B entering Tank A when inventory, concentration, and dilution change the credible mechanism.

Term / Concept	Definition
Chemical incompatibility screening	A structured review of what could happen if two chemicals are accidentally mixed, such as heat generation, gas evolution, toxic vapor release, fire, pressure rise, or corrosion.
C1-C6 severity scale	A consequence ranking scale used in the screening workbook. C1-C4 represent lower to moderate consequences; C5-C6 represent high-severity outcomes that typically require stronger safeguards or review.
C5/C6	The high-severity band. In this project, C5 and C6 are often treated similarly from a mitigation trigger standpoint, so a C6-to-C5 change may not materially change the required safeguard.
RTD	Resistance Temperature Detector. A sensor installed in a tank to measure temperature. It is useful only when the hazardous event produces a detectable bulk temperature rise in time for action.
CAMEO-style hazard flags	Conservative theoretical indicators such as heat generation, gas generation, toxic gas, fire, or explosive potential due to the reaction of two chemicals. These flags are useful for screening but can overstate what is physically credible in a real process setting.
Gold-set audit	A manually reviewed subset of chemical pairs used to check whether the weak labeling rules agree with expert engineering judgment.
Weak supervision / labeling functions	A method for creating training labels from multiple partial rules. Each rule votes only when its conditions apply; otherwise it abstains. The votes are aggregated into a weak label for model training.
Pair-level vs. tank-level decision	The dataset evaluates chemical pairs, but the physical RTD installation decision is made for a storage tank that normally contains one chemical. The system therefore aggregates pair-level predictions into a tank-level recommendation.
Directional scenario	A tank-context scenario that defines which chemical is stored as the majority inventory and which incompatible chemical enters as the abnormal contaminant, such as

	A_into_B or B_into_A.
Inventory context	Tank level, stored quantity, concentration, dilution, and order-of-addition information that can affect heat release, gas evolution, kinetics, and whether an RTD detects a useful bulk temperature signal.
Review routing	A conservative decision state used when the model or labeling functions do not have enough evidence to force a binary RTD recommendation.

1. Business Problem & Context

Chemical manufacturing sites intentionally use conservative screening methods when performing process and project hazard analyses because the cost of missing a severe incompatibility can be high. However, conservative screening can also create over-scoped mitigation designs, repeated lab testing, and decision fatigue when every high-severity flag is treated as if it requires the same hardware solution.

The original project hypothesis was that conservative screening would frequently overstate severity. The completed gold-set audit showed a more nuanced result: most severity ratings remained unchanged or moved within the same high-severity band, such as C6 to C5. Because C5 and C6 both generally trigger mitigation review, those changes do not always reduce scope.

The stronger and more actionable problem is mitigation selection. A scenario may truly be high severity, but an RTD may still be the wrong safeguard if the dominant hazard is toxic gas, pressure generation, fire, very fast decomposition, or a gel/sludge/fouling mechanism that prevents reliable temperature detection.

A related asset-level issue is directionality: the same pair can have different practical consequences depending on which material is stored in the tank, which material is introduced, and the tank inventory at the time of the abnormal transfer. The MVP therefore treats directionality as a tank-scenario and review-routing layer rather than hiding it inside the pair-level classifier.

Original Framing	Revised Framing	Why the Revision Matters
Can ML reduce conservative severity ratings?	Can ML identify when RTD installation is actually useful, creditable as a safeguard, and practical?	This aligns the model target with the observed data signal and with a real engineering decision that affects capital scope.

2. System Architecture

The system follows a piped-inference architecture. In plain language, this means the system transforms structured chemical-pair data into intermediate mechanism signals, aggregates weak supervision votes, trains a tabular model, and then converts pair-level predictions into direction-aware, tank-level recommendations.

The Phase 1 MVP is intentionally tabular and interpretable. It uses the public structured dataset, gold-set audits, calibrated weak-labeling functions, leakage-safe baseline models, and a directionality-aware tank-scenario layer to build a credible proof of concept. Phase 2, which would use SDS/NLP extraction and richer site data, remains future scope.

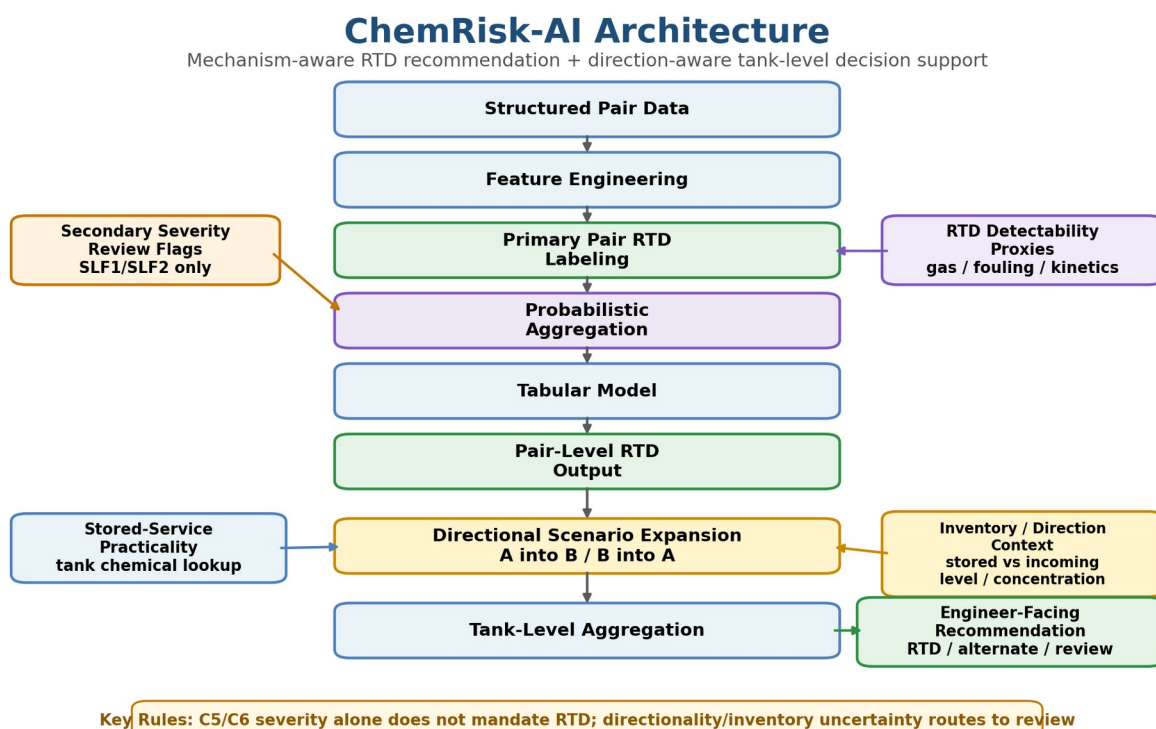


Figure 1. Revised ChemRisk-AI architecture. The primary target is pair-level RTD recommendation; severity demotion remains a sparse review flag; directionality and inventory context are handled as a tank-scenario layer before final asset-level recommendations.

2.1 Phase 1 Objective - Capstone MVP

Develop a reliable, interpretable, tabular decision-support model that:

- Predicts pair-level RTD usefulness / RTD required vs. not required.
- Explains the dominant control failure mode when RTD is not appropriate.
- Provides severity context and rule-based flags for the small subset of decision-relevant severity demotion candidates below C5.
- Aggregates pair-level predictions to direction-aware tank-level RTD recommendations using stored chemical service practicality and A_into_B / B_into_A scenario logic.
- Flags directionality-sensitive or inventory-dependent cases for engineering review when tank level, concentration, or order-of-addition could change the credible mechanism.
- Quantifies confidence and routes uncertain cases to peer review or lab validation.

2.2 Phase 2 Future Extension

Phase 2 would expand the system beyond known structured rows by extracting chemical features from Safety Data Sheets (SDSs). This could include incompatible materials, decomposition products, pH, and physical-property clues. This extension is useful for future generalization, but it is not required for the Phase 1 capstone MVP.

3. Data Readiness & Current Dataset Reality

The current public dataset is sufficient for a weak-supervised, tabular MVP focused on RTD recommendation and mitigation selection. It is not sufficient for a full mechanistic chemistry predictor, a robust six-class severity model, or a complete kinetics model that resolves direction, concentration, and tank inventory without site context. This distinction is important because it keeps the project credible.

Item	Current Status / Interpretation
Rows available	Approximately 313 public chemical-pair rows.
Gold-set status	21 audit rows completed for first-pass LF development and calibration. A frozen v1.1 prospective audit batch of 15 additional rows was selected using model disagreement, uncertainty, and mechanism-coverage criteria. The full 50-row target remains useful for stronger validation.
Strongest signal	RTD usefulness / mitigation misapplication.
Weakest signal	Full six-class severity prediction and severity-demotion model training.
Main limitation	No direct measured kinetics, heat of reaction, viscosity, phase behavior, tank level, concentration, order-of-addition, or incident outcome data.
Current implementation status	Python helper features and RTD LFs were rebuilt and matched to the spreadsheet snapshot. A calibrated v1.1 weak-label layer and leakage-safe baseline models have been implemented in Colab.
Directionality status	Directionality is now handled as a separate scenario-expansion and review-routing layer, not as an unsupported claim inside the pair-level model.

3.1 What the 21-Row Gold-Set Audit Established

- Six of twenty-one severity ratings changed in the audit.
- Four of those six changed from C6 to C5, which usually does not change the mitigation trigger because both remain in the high-severity band.
- Two changed from C6 to C2, which is decision-relevant. Those cases were associated with thermal buffering/dilution and nonhazardous exotherm behavior.
- The dominant value opportunity is not broad severity demotion. It is identifying whether RTD installation is useful, ineffective, or the wrong safeguard type.

3.2 Recent Severity-Demotion Finding

The severity-demotion signal is real but sparse. In the public workbook, the active severity-demotion flags currently produce only a small number of candidate rows. That is useful for triage and engineering review, but not enough to justify a separate supervised severity-demotion model in the Phase 1 MVP.

3.3 Frozen v1.1 and Prospective Audit Cycle

After spreadsheet-to-Python parity was achieved, targeted gold-set diagnostics exposed a small number of RTD false positives and one false negative. These were addressed in a calibrated v1.1 weak-label layer without overwriting the spreadsheet-parity baseline. The 21-row audit set remains a development/calibration set because it informed the v1.1 adjustments.

The next 15 audit candidates were selected after freezing v1.1 using active-learning criteria: model disagreement, out-of-fold uncertainty, moderate-confidence single-LF votes, minority-class RTD-required cases, and mechanism coverage gaps. This prospective audit cycle is a stress test of v1.1, not a random population-level accuracy estimate.

4. Weak Supervision Framework

Because high-consequence incident data is sparse, the project uses weak supervision instead of traditional supervised learning with clean ground-truth labels. Labeling functions are engineering-informed rules that vote only when their conditions apply. If a rule does not have enough evidence, it abstains rather than forcing a vote.

Abstentions are ignored during aggregation. For example, if three active rules vote [1, 0, 1] for RTD installation and five other rules abstain, the unweighted probability of RTD required is 2/3. Abstentions are not averaged as negative values.

4.1 Secondary Severity-Demotion Review Flags

Severity calibration is retained only as secondary context. It is not the main model target. The current workbook treats severity demotion as a sparse rule-based review flag rather than a supervised learning target.

Severity LF	Current v1 Status	Use in Project
SLF1_ThermalBuffer_Demotion	Candidate review flag. Broad proxy: high severity plus diluent / thermal-buffer indication. Current public workbook restores this as a candidate flag, not a validated model target.	Flags potential cases where dilution or thermal mass may make the initial severity overly conservative.
SLF2_NonHazardousExotherm_Demotion	High-value candidate review flag. Current proxy identifies sludge/gel/polymer/precipitation-style nonhazardous-exotherm patterns. Low sample size but validated on the limited audited rows where it voted.	Flags cases where heat may be present but not hazardous or runaway-capable.
SLF3_OpenAirExposure_Demotion	Retired from active v1 aggregation due to poor early gold-set precision.	Mentioned only as future work; not used in the current workbook or Colab target.

These severity-demotion flags should be reported in Colab as candidate review opportunities. They should not be used to train a separate severity-demotion classifier unless more positive examples are audited later.

4.2 Primary Mitigation-Decision Layer

The primary layer predicts whether an RTD is required and creditable as a thermal safeguard. The active labeling function (LF) system intentionally separates the broad baseline rule from the final vote. The broad rule votes for RTD installation for C5/C6 reactions that generate heat, but the rule is retained only as a comparator because the gold-set audit showed it over-recommends RTDs.

Labeling Function	Definition
LF1_NoHeat_LowSeverity_NoRTD	Votes no RTD when the row is below the high-severity band or has no heat-generation flag.
LF2_BulkLiquidNeutralization_RTD	Votes RTD required for clean, high-severity bulk-liquid acid/base exotherms with no obvious fouling, gas, or gel proxy.
LF3_ToxicGasDominant_NoRTD	Votes no RTD when the likely dominant hazard is toxic gas or vapor release, not useful bulk temperature rise.
LF4_DetectabilityFailure_NoRTD	Votes no RTD when classes imply gel, sludge, precipitation, phase separation, gas shrouding, or fouling.

LF5_FastPressureDecomp_NoRTD	Votes no RTD when fast gas/pressure/decomposition mechanisms are likely to outrun RTD-based prevention.
LF6_OxidizerOrganic_NoRTD_Candidate	Candidate no-RTD vote for oxidizer-organic/fire/decomposition patterns; requires continued gold-set validation.
LF7_Baseline_SevereHeat_RTD	Baseline comparator only. It is excluded from final weak voting because it is too broad.
LF9_Thermal_Default_RTD	Controlled positive RTD pathway: severe + heat + no active negative failure-mode proxy. In v1.1, LF9 is recomputed after calibrated negative rules are applied.
LF8_Diluent_NoRTD_Candidate	Candidate no-RTD vote for severe heat rows where dilution / thermal buffering and absence of toxic gas or explosive potential suggest the exotherm may not require RTD as the primary safeguard.
LF10_OxidizingAcidChelant_NoRTD	Calibrated v1.1 no-RTD vote for oxidizing acid + chelant scenarios where gas, pressure, or wrong-hazard-type behavior dominates over useful bulk temperature detection.
LF11_AcidicSurfactantGlycolEther_NoRTD	Calibrated v1.1 no-RTD vote for acidic anionic surfactant + glycol ether solvent cases where detectability or wrong-safeguard concerns are more credible than an RTD-credible bulk thermal event.
LF12_AcidicSurfactantVolatileBase_RTD	Calibrated v1.1 positive RTD vote for severe heat-generating acidic anionic surfactant + volatile/ammoniacal base neutralization where the audited mechanism supports an RTD-credible thermal safeguard.

4.3 Probabilistic Aggregation

- Collect all nonblank votes from the active LF set.
- Ignore abstentions.
- Compute `weak_vote_prob` as the mean or weighted mean of active votes.
- Assign RTD required, RTD not required, or Review based on probability and conflict rules.
- Measure LF performance against the audited gold set using coverage, precision when voting, and conflict rate.

4.4 Calibrated v1.1 Weak-Label Layer

The Colab notebook preserves the original spreadsheet-parity LF reconstruction using `py_lf...` columns, then creates a separate calibrated v1.1 layer using `cal_lf...` columns. This versioning is intentional: parity proves reproducibility, while v1.1 documents targeted mechanism corrections learned from the development audit set.

The v1.1 layer should be frozen before reviewing prospective audit rows. If the next audit cycle reveals new failure modes, those corrections should be versioned as v1.2 rather than silently changing v1.1. This protects the capstone from appearing to tune rules until they match the answer.

5. Mitigation Feasibility / RTD Detectability

An RTD is not automatically useful just because a reaction generates heat. It is useful only when the event produces a bulk temperature signal that is detectable in time for action and the stored material does not make the instrument unreliable.

The Phase 1 dataset does not contain direct measured viscosity, mixing quality, heat-transfer coefficient, thermowell fouling, bulk-vs-localized temperature response, tank level, concentration, or direction/order-of-addition data for every chemical pair. Therefore, the system uses engineered proxy indicators derived from chemical class, reaction type, hazard flags, and a separate tank-scenario review layer.

The Phase 1 RTD detectability proxies include:

- Bulk-liquid exotherm proxy:

inferred from reaction type, heat-generation flag, high-severity context, and chemical-class combinations consistent with clear-liquid acid/base exotherms.

- Detectability-failure proxy:

inferred from chemical-class combinations associated with gel formation, sludge, precipitation, phase separation, polymer/emulsion instability, or thermowell fouling.

- Gas / pressure / sensor-lag proxy:

inferred from gas-generation, explosive-potential, oxidizer/decomposition chemistry, and sparse reaction-speed information where available.

- Wrong-hazard-type proxy:

inferred when toxic gas, fire, pressure, corrosion, or decomposition appears to dominate over bulk thermal runaway.

- Stored-service RTD practicality:

determined using a separate chemical-service lookup that flags polymer, emulsion, wax, slurry, silicate, or other fouling-prone tank services.

- Directionality / inventory sensitivity proxy:

inferred when acid/base neutralization, oxidizer/reducer chemistry, hypochlorite/acid or amine chemistry, silicate gel formation, polymer/emulsion destabilization, metal complexation, gas generation, or unknown reaction speed could make A_into_B different from B_into_A.

- Valid RTD-use proxy:

inferred when a credible severe heat-generating scenario is present and no toxic-gas, detectability-failure, fast-pressure, or stored-service impracticality proxy dominates.

5.1 RTD Recommendation Logic

RTD is recommended only if all of the following are true:

- A credible exothermic reaction is present.
- The exotherm is hazardous, not merely heat-generating.
- The event is expected to produce a bulk temperature rise detectable by RTD.
- No dominant failure mode makes RTD ineffective, such as toxic-gas dominance, gas shrouding, gel/sludge/fouling, phase separation, or reaction kinetics too fast for preventive action.
- At the tank level, the stored chemical service is RTD-practical and creditable for the relevant direction of the abnormal transfer.
- Directionality-sensitive or inventory-dependent cases are routed to engineering review unless sufficient site context resolves the stored/incoming-material scenario.

If any of these conditions fail, RTD is not recommended as the primary thermal safeguard. Other safeguards may still be required.

6. Tank-Level RTD Aggregation Layer

The source data is chemical-pair based, but RTD installation is an asset-level decision. A tank normally stores one chemical service, while the incompatible chemical is introduced by an abnormal transfer or unloading scenario. Therefore, pair-level model output must be expanded into direction-aware tank scenarios before final recommendations are made.

The tank-level decision combines three inputs:

- Pair-level weak/model prediction: whether the pair has a credible RTD-creditable thermal mechanism at the screening level.
- Stored-service RTD practicality lookup: whether the chemical normally stored in the tank is compatible with reliable temperature measurement.
- Aggregation across all relevant incompatible pairs and directions for that stored chemical, including A_into_B and B_into_A scenario expansion where appropriate.

For example, a polymer or wax emulsion tank may be RTD-impractical even if one pair-level scenario includes heat, because normal service fouling or coating could make the RTD unreliable. Similarly, an acid/base or oxidizer/reducer pair may need directionality review because the mechanism can depend on which material is present as the majority tank inventory. In those cases, the system recommends alternate safeguards or engineering review rather than crediting RTD as the primary safeguard without site context.

6.1 Directionality and Inventory Context

Directionality is handled as a scenario layer instead of being embedded as an unsupported assumption in the pair-level classifier. For each incompatible pair, the tank-level layer can create two conceptual scenarios:

A_into_B, where chemical B is the stored/major inventory and A is the incoming abnormal contaminant; and B_into_A, where chemical A is stored and B enters abnormally.

This matters because majority/minority material, tank level, concentration, dilution, heat capacity, local mixing, and pH endpoint can change the credible mechanism. A reaction that is RTD-credible when a small amount of acid enters a large base tank may not behave the same way when the order is reversed, especially for gas evolution, fast decomposition, precipitation, gel formation, or localized exotherms.

In Phase 1, the public dataset does not contain enough site-specific information to calculate these effects. The model therefore uses conservative directionality-sensitive flags and review routing. If either direction appears RTD-credible, the case remains a candidate RTD scenario. If either direction is materially direction-sensitive and unresolved, the final tank recommendation should include engineering review rather than a fully automated binary conclusion.

7. Modeling Plan for Colab Implementation

The Colab implementation should follow the documented dataset and avoid relying on Excel recalculation. The notebook should rebuild the features and LFs in Python so the workflow is reproducible, then layer calibrated labels, baseline models, prospective audit selection, and direction-aware tank-scenario logic on top of the verified foundation.

1. Load the public data from the Public_Colab_Values-style export or CSV.
2. Validate schema, data types, and missing values.
3. Rebuild helper features and mechanism-aware labeling functions in Python.
4. Compare Python LF results to the spreadsheet snapshot as a QA check.
5. Evaluate LF coverage, precision when voting, and conflict rate using the 21 completed development audit rows.
6. Freeze calibrated v1.1 weak labels, train leakage-safe baseline tabular models on cal_weak_rtd_label, and treat Review cases separately or as uncertainty cases.
7. Do not train a separate severity-demotion model in v1; report severity-demotion flags as candidate engineering review opportunities.
8. Validate pair-level predictions against audited rtd_required and control_failure_mode where available, clearly separating development audits from prospective audits.
9. Use model disagreement and near-boundary scores to select prospective audit rows, then evaluate frozen v1.1 before considering v1.2 rule changes.
10. Demonstrate direction-aware tank-level aggregation with stored-service RTD practicality and A_into_B / B_into_A scenario expansion.

8. Evaluation & Validation Strategy

Evaluation must match the revised capstone objective. The most important question is not whether the model can perfectly reproduce a six-class severity label. The most important question is whether the model produces useful, conservative, and explainable mitigation recommendations.

Metric	Definition
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LF coverage	How often each LF produces a nonblank vote.
LF precision when voting	How often a rule agrees with the audited target when it actually votes. Abstentions are not counted as wrong.
LF conflict rate	How often active LFs disagree on the same row.
RTD recommendation precision	How often predicted RTD decisions agree with audited rtd_required decisions.
Review routing	Whether uncertain/conflicting cases are sent to peer review rather than forced into false certainty.
Tank-level practicality	Whether tank recommendations respect stored-service RTD feasibility.
Severity-demotion flag quality	Reported as candidate-review performance only; not treated as a separate supervised model target in v1.
Prospective audit performance	Agreement of frozen v1.1 weak labels and model predictions against audit rows selected after the rules were frozen. Reported separately from development-set performance.
Directionality review routing	Whether A_into_B / B_into_A or inventory-sensitive cases are correctly routed to tank-level review when site context is insufficient.
Class-imbalance metrics	Balanced accuracy, RTD precision, RTD recall, F1 score, average precision, and false negatives are emphasized over plain accuracy.

9. Business Impact

The revised business case is mitigation optimization, not blanket severity reduction. This is more credible because the gold-set audit showed severity is often correct, while RTD applicability varies materially by mechanism and detectability.

- OpEx reduction: avoid unnecessary lab testing for high-confidence, well-understood cases.
- CapEx optimization: avoid blanket RTD installation where RTDs are not creditable or not the correct safeguard.
- Decision standardization: encode repeatable mechanism-aware engineering logic into a transparent framework.
- Portfolio prioritization: support future ranking of safeguards by risk reduction per dollar.
- Scenario realism: avoid one-size-fits-all recommendations by distinguishing pair-level chemical incompatibility from tank-level direction, inventory, and stored-service practicality.

Screening-level savings estimates should be presented as directional. The model does not authorize safety scope reduction by itself; it provides auditable evidence for engineering review and detailed design.

10. Tech Stack

Tool / Method	Role in Project
Python / Pandas / NumPy	Data processing, feature engineering, LF implementation, and analysis.
Scikit-learn / XGBoost	Baseline and tree-based tabular modeling.
SHAP	Interpretable feature attribution for engineer-facing explanations.
Snorkel-style weak supervision	Optional formalization of LF voting and probabilistic label aggregation.
Streamlit / CLI (future)	Demonstration interface for engineers.
spaCy / Transformers (future)	Phase 2 SDS entity extraction, not part of the Phase 1 MVP.
OpenPyXL / Excel audit templates	Maintaining auditable gold-set workbooks, formula-driven audit fields, and prospective audit candidate lists.

11. Roadmap

1. Finalize public dataset and gold-set mapping fields.
2. Implement Python LF framework and verify spreadsheet-to-Python agreement. Completed for helper features and RTD LFs.
3. Run EDA on severity, hazard flags, LF coverage, weak labels, review rates, and severity-demotion candidate flags.
4. Train leakage-safe baseline models for calibrated pair-level RTD weak labels using grouped cross-validation.
5. Evaluate calibrated v1.1 against the 21 completed development audit rows, while documenting that this is not independent validation.
6. Use model disagreement, low-confidence cases, and mechanism gaps to choose the next 10-15 prospective audit rows. Completed for the frozen v1.1 audit cycle.
7. Audit the prospective v1.1 candidate rows, evaluate frozen v1.1 performance, and only then decide whether v1.2 rule changes are justified.
8. Add directionality and inventory-context scenario expansion before final tank-level recommendation outputs.
9. Build README, architecture explanation, directionality/tank-level demonstration, and portfolio-ready outputs for GitHub.

12. Failure Modes & Limitations

- The model is trained on weak labels, not incident data or laboratory reaction-test outcomes.
- The current gold-set is useful for MVP calibration but is still limited in size.
- Proxy features cannot replace measured kinetics, heat of reaction, viscosity, phase behavior, tank level, concentration, order-of-addition, or mixing data.
- Chemical-class rules may miss edge cases or overgeneralize within a class.
- Severity-demotion positives are sparse; severity-demotion outputs should be treated as review flags, not a mature predictive model.
- Tank-level practicality is a lookup-based engineering proxy and requires site validation before use in real design.
- The pair-level model does not by itself resolve direction of reaction. Directionality is handled downstream through scenario expansion and review routing, and final design requires site context.
- The system must remain human-in-the-loop for safety-critical decisions.

13. Conclusion

ChemRisk-AI is best framed as a mechanism-aware safeguard decision-support system. It does not claim to predict chemical truth from limited data or to solve reaction kinetics without site context. Instead, it uses weak supervision, gold-set audit evidence, interpretable tabular modeling, prospective audit selection, and

direction-aware tank-level aggregation to support a specific engineering decision: whether RTD installation is useful, creditable, and practical.

Severity demotion remains valuable when it is credible, but the current evidence supports using it as a sparse candidate review flag rather than a full model target. The directionality layer strengthens the capstone because it explicitly recognizes the difference between a pair-level incompatibility screen and an asset-level abnormal-transfer scenario. This framing is stronger for industrial AI roles because it demonstrates domain expertise, realistic treatment of data limitations, transparent weak supervision, interpretable modeling, and business impact tied to a real engineering workflow.

Appendix A. Revision Log - What Changed and Why

This log is placed after the main document so readers first understand the problem, architecture, and decision logic before reviewing how the project evolved.

Revision	Reason
Moved revision log to appendix	Domain context should come before change history so readers are not confused by project-specific terms.
Added reader orientation	Defines C1-C6, C5/C6, RTD, weak supervision, labeling functions, gold set, and pair vs tank decisions for readers without chemical-process background.
Reframed primary target	Shifted from full severity prediction to pair-level RTD usefulness and tank-level RTD recommendation.
Clarified severity-demotion role	Severity demotion is now a secondary rule-based review flag, not a primary model target. SLF1 and SLF2 remain candidate flags; SLF3 is retired from v1.
Clarified dataset readiness	Current data supports a weak-supervised tabular MVP, not a first-principles chemistry predictor.
Added tank-level aggregation	Explains how pair-level predictions become storage-tank RTD recommendations using stored-service practicality.
Updated architecture diagram	Diagram now shows severity as sparse review flags, RTD detectability as primary weak-supervision target, and tank-level aggregation as the final engineering decision layer.
Added calibrated v1.1 weak-label layer	Documents the separation between spreadsheet-parity LF reconstruction and targeted v1.1 calibration after development audit diagnostics.
Added prospective audit cycle	Captures that the next 15 audit rows are selected after v1.1 is frozen using model disagreement, uncertainty, and mechanism coverage gaps.
Added directionality and inventory-context layer	Addresses the engineering limitation that A_into_B can differ from B_into_A because stored inventory, dilution, concentration, and order-of-addition can affect mechanism and RTD detectability.
Updated modeling and evaluation plan	Adds leakage-safe grouped cross-validation, class-imbalance metrics, evidence-tier weighting, prospective audit performance, and directionality review routing.